

AI Engineer Resume Tips & Portfolio

Projects (2026)

SUMMARY

A strong AI Engineer resume should demonstrate your ability to build, deploy, and maintain real-world AI systems—

not just train machine learning models. Recruiters look for practical experience with Machine Learning, Deep

Learning, Generative AI, MLOps, cloud deployment, and software engineering principles. A portfolio containing

production-ready AI applications is often more valuable than numerous certifications.

PROJECTS

House Price Prediction System

- Skills
- Scikit-learn
- Regression
- FastAPI
- Docker

Medical Image Classification

- Skills
- CNN
- PyTorch
- Computer Vision
- Transfer Learning

Spam Email Detection

- Skills
- NLP
- Text Classification
- Scikit-learn

Customer Churn Prediction Dashboard

- Skills
- Machine Learning
- Data Visualization
- Streamlit

Face Mask Detection

- Skills
- OpenCV
- YOLO
- Object Detection

AI Resume Screening System | Extract resumes, rank candidates, and generate recruiter summaries.

- Skills
- NLP
- OCR
- LLM
- FastAPI

Intelligent PDF Question Answering | Upload PDFs and answer questions using RAG.

- Skills
- LangChain
- Vector Database
- Embeddings
- Retrieval

AI Customer Support Chatbot | Answer company-specific questions using internal documents.

- Skills
- RAG
- Prompt Engineering
- LangGraph
- Astra DB

AI Meeting Summarizer | Generate meeting summaries, action items, and follow-up tasks from transcripts.

- Skills
- Whisper
- LLM
- NLP

Skin Disease Detection System | Detect skin diseases from images and provide symptom-based recommendations.

- Skills
- YOLO
- CNN
- MLflow
- FastAPI

Multi-Agent Research Assistant ■ | Create specialized AI agents for planning, searching, summarizing, fact-checking, and report generation.

- Skills
- LangGraph
- Multi-Agent Systems
- LLMs

- RAG

Enterprise Knowledge Assistant ■ | Build an AI assistant capable of answering questions from thousands of

enterprise documents.

- Skills
- Hybrid Search
- Vector Database
- Metadata Filtering
- Semantic Search
- Citation Generation

AI Code Review Assistant | Automatically review GitHub pull requests and suggest improvements.

- Skills
- LLM
- GitHub API
- Prompt Engineering

AI Medical Assistant | Analyze medical reports, answer patient queries, and generate simplified explanations while

citing supporting medical documents.

- Skills
- RAG
- Medical NLP
- OCR
- LangGraph

AI Financial Report Analyzer | Extract insights, summarize financial statements, compare companies, and answer

investment-related questions.

- Skills
- Document Parsing
- LLM
- RAG
- Visualization

Autonomous Data Analysis Agent | Upload CSV or Excel files and allow an AI agent to perform exploratory data

analysis, generate charts, answer questions, and produce reports.

- Skills
- Python
- Pandas
- LLM
- Agentic AI

AI Learning Assistant | Generate personalized learning roadmaps, quizzes, flashcards, and progress tracking for any

technical subject.

- Skills
- LangGraph
- Memory
- RAG
- PDF Generation

AI Legal Document Assistant | Summarize legal documents, identify clauses, compare contracts, and answer legal

questions using retrieved evidence.

- Skills
- NLP
- RAG
- Vector Database
- LLM

Multi-Modal AI Assistant ■ | Develop an assistant capable of understanding text, images, PDFs, and audio while

generating intelligent responses.

- Skills
- Vision Models
- OCR
- Speech-to-Text
- LLM
- LangGraph

INTRODUCTION

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Learning, Generative AI, MLOps, cloud deployment, and software engineering principles. A portfolio containing

production-ready AI applications is often more valuable than numerous certifications.

RESUME TIPS

1. Keep Your Resume ATS-Friendly: Use a clean, simple format that is easy for Applicant Tracking Systems (ATS) to

parse. Include standard sections: Contact Information, Professional Summary, Technical Skills, Projects, Experience (if

any), Education, Certifications, Achievements. Keep your resume to one page if you are a student or have less than

three years of experience.

2. Write a Strong Professional Summary: Briefly describe your background, technical expertise, and career objective.

Example: Aspiring AI Engineer with hands-on experience in Machine Learning, Deep Learning, Large Language

Models (LLMs), Retrieval-Augmented Generation (RAG), LangGraph, and MLOps. Passionate about building

scalable AI applications and deploying production-ready machine learning systems.

3. Organize Technical Skills: Programming: Python, SQL. Machine Learning: Scikit-learn, XGBoost, LightGBM.

Deep Learning: PyTorch, TensorFlow, Keras. Generative AI: LangChain, LangGraph, LlamaIndex, Prompt

Engineering, RAG. LLM APIs: OpenAI, Groq, Google Gemini, Anthropic Claude. Vector Databases: Astra DB,

Pinecone, Qdrant, ChromaDB. MLOps: MLflow, Docker, FastAPI, GitHub Actions. Cloud: AWS, Azure, Google

Cloud.

4. Make Projects the Highlight: For every project include Problem Statement, Technologies Used, AI Techniques Used,

Key Features, Deployment Link, GitHub Repository, Results or Impact.

5. Quantify Results: Examples include 94% classification accuracy, 35% reduced latency, 100,000 knowledge chunks

indexed, 500+ daily users.

6. Demonstrate End-to-End AI Skills: Data Collection, Data Processing, Model Training, Model Evaluation, API

Development, Deployment, Monitoring, Documentation.

7. Include GitHub & Portfolio: GitHub Profile, LinkedIn Profile, Portfolio Website, optional Technical Blog.

Repositories should include README, Architecture Diagram, Installation Guide, API Documentation, Screenshots,

optional Demo Video.

8. Highlight MLOps Experience: MLflow, Docker, Model Versioning, Experiment Tracking, CI/CD, Cloud

Deployment.

9. Optimize for Keywords: Machine Learning, Deep Learning, CNN, Transformers, LLM, RAG, LangChain,

LangGraph, Vector Database, MLOps, Docker, FastAPI, MLflow, AWS.

10. Proofread Before Applying: Ensure no spelling mistakes, broken links, or inconsistent formatting; verify listed

skills can be discussed in interviews.

PROJECTS THAT IMPRESS RECRUITERS THE MOST

If you can only build five projects, prioritize these: 1) AI Interview Preparation Platform, 2) Enterprise Knowledge

Assistant, 3) Multi-Agent Research Assistant, 4) Skin Disease Detection System (End-to-End), 5) Autonomous Data

Analysis Agent. These projects showcase expertise across Machine Learning, Deep Learning, Generative AI, RAG, AI

Agents, MLOps, deployment, and software engineering.

FINAL ADVICE

Employers are looking for AI Engineers who can build complete, production-ready systems rather than isolated

machine learning models. Focus on creating projects that demonstrate data preprocessing, model development,

Retrieval-Augmented Generation (RAG), Large Language Model integration, API development, cloud deployment,

monitoring, and documentation. A portfolio containing three to five polished, well-documented projects with live

demos, GitHub repositories, and clear architecture diagrams will make your resume significantly more competitive in

the 2026 AI job market.